## CONTROL SYSTEM DEVICE AND CONTROL RELATIONSHIP CHART [SKYACTIV-D 1.5]

id0103q2800200

×: Applicable

| Item | MAIN RELAY CONTROL | IDLE SPEED CONTROL | BOOST CONTROL | FUEL INJECTION AMOUNT CONTROL | MULTIPLE FUEL INJECTION CONTROL | FUEL INJECTION TIMING CONTROL | FUEL PRESSURE CONTROL | INTAKE SHUTTER VALVE CONTROL | EXHAUST SHUTTER VALVE CONTROL | COOLANT CONTROL VALVE CONTROL | ELECTRIC WATER PUMP CONTROL | GLOW CONTROL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Input device | | | | | | | | | | | | |
| Check connector | | | | | × | | | | | | | |
| IG1 relay | × | | × | × | × | × | | × | × | × | × | × |
| Brake switch (No.1 signal) | | | | | | | | | | | | |
| Brake switch (No.2 signal) | | | | | | | | | | | | |
| MAP sensor | | × | × | × | × | × | | × | | | | × |
| Boost air temperature sensor | | | | | | | | | | | × | |
| MAF sensor | | × | × | × | × | × | × | × | × | | | |
| IAT sensor No.1 | | × | × | × | × | × | × | × | × | × | × | × |
| IAT sensor No.2 | | | | × | × | × | × | × | × | | × | |
| ECT sensor | | × | × | × | × | × | × | × | × | × | | × |
| Fuel temperature sensor | | | | × | × | × | × | | | | | |
| APP sensor No. 1, No.2 | | × | × | × | × | × | × | × | × | × | × | × |
| Fuel pressure sensor | | × | | × | × | × | × | | | | | |
| Intake shutter valve position sensor | | × | | | | | | × | | | | × |
| Exhaust shutter valve position sensor | | | | | | | | | × | | | |
| Coolant control valve position sensor | | | | | | | | | | × | | |
| HP-EGR control valve position sensor | | × | × | | | | × | | | | | × |
| LP-EGR control valve position sensor | | | × | | | | × | | | | | × |
| Exhaust gas pressure sensor No.1 | | | × | × | × | × | × | | | | | |
| Exhaust gas pressure sensor No.2 | | | | × | × | × | × | | | | | |
| BARO sensor | | × | × | × | × | × | × | × | × | | | × |
| Battery | | | × | | | | | | | × | × | × |
| Refrigerant pressure sensor | | × | | | | | | | | | | |
| Exhaust gas temperature sensor No.1 | | | × | × | × | × | × | | | | | |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Exhaust gas temperature sensor No.2 | | | | × | × | × | × | | | | | |
| Exhaust gas temperature sensor No.3 | | | | × | × | × | × | | | | | |
| Exhaust gas temperature sensor No.4 | | | × | | | | | | × | | | |
| Engine oil temperature sensor | | | | | | | | | | × | | |
| Engine oil pressure sensor | | × | | | | | | | | | | |
| Actuator position sensor (turbocharger with variable turbine geometry) | | | × | | | | | | | | | |
| Speed sensor assy (turbocharger with variable turbine geometry) | | | × | × | | | | | | | | |
| Ambient temperature sensor | | | | | | | | | | | | |
| A/F sensor | | | | × | × | × | × | | | | | |
| Power brake unit vacuum sensor | | | | | | | | | | | | |
| CKP sensor | × | × | × | × | × | × | × | × | × | × | × | × |
| CMP sensor | | × | | × | × | × | × | × | × | | × | |
| Back-up light switch (MTX) | | × | | × | × | × | | | | | | |
| Neutral switch No.1 (MTX) | | × | × | × | × | × | | | | | | |
| CPP switch (MTX) | | × | | × | × | × | | × | × | | | |
| Starter interlock switch (MTX) | | | | × | × | × | | | | | | |
| Neutral switch No.2 (MTX) | | | | × | × | × | | | | | | |
| Clutch stroke sensor (MTX) | | × | | | | | | | | | | |
| Generator (stator coil) (without i-ELOOP) | | | | × | × | × | × | | | | | |
| CAN*1 | | × | × | × | × | × | × | × | × | × | × | × |
| LIN*2 | | | | | | | | | | | | × |
| Output device | | | | | | | | | | | | |
| Intake shutter valve | | × | | | | | | × | | | | |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Exhaust shutter valve | | | | | | | | | × | | | |
| HP-EGR control valve | | × | | | | | | | | | | |
| LP-EGR control valve | | | | | | | | | | | | |
| Fuel pressure relief valve | | | | | | | × | | | | | |
| Main relay | × | | | | | | | | | | | |
| Turbocharger solenoid valve | | | × | | | | | | | | | |
| A/F sensor heater | | | | | | | | | | | | |
| Starter relay | | | | | | | | | | | | |
| DC-DC converter (without i-ELOOP) | | | | | | | | | | | | |
| Generator (Field coil) (without i-ELOOP) | | | | | | | | | | | | |
| A/C relay | | | | | | | | | | | | |
| Fan control module | | | | | | | | | | | | |
| Cooling fan relay | | | | | | | | | | | | |
| Engine oil solenoid valve | | × | | | | | | | | | | |
| Suction control valve | | × | | | | | × | | | | | |
| Fuel injector No. 1—No.4 | | × | | × | × | × | | | | | | |
| Blow-by heater relay | | | | | | | | | | | | |
| Electric water pump | | | | | | | | | | | × | |
| Coolant control valve | | | | | | | | | | × | | |
| LIN*2 | | | | | | | | | | | | × |
| CAN*3 | | | | | | | | | | | | × |

*1 : DSC HU/CM, TCM (ATX), EPS control module, SAS control module, instrument cluster, BCM, start stop unit

*2 : Glow control module, engine oil level sensor

*3 : Instrument cluster (charging system warning light, master warning light, glow indicator light, diesel particulate filter indicator light, i-stop warning light (amber)/i-stop indicator light (green), i-stop indicator alarm/i-stop warning alarm)

×: Applicable

| Item | EGR CONTROL | AIR FUEL RATIO (A/F) SENSOR HEATER CONTROL | DIESEL PARTICULATE FILTER REGENERATION CONTROL | A/C CUT-OFF CONTROL | ELECTRICAL FAN CONTROL | STARTER CUT-OFF CONTROL | GENERATOR CONTROL | RUN DRY PREVENTION (RDP) CONTROL | BLOW-BY HEATER CONTROL | i-stop CONTROL | DC-DC CONVERTER CONTROL | ENGINE OIL CONTROL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Input device | | | | | | | | | | | | |
| Check connector | | | × | | | | | | | | | |
| IG1 relay | | | | × | × | × | | | | × | × | |
| Brake switch (No.1 signal) | | | | × | | | | | | × | | |
| Brake switch (No.2 signal) | | | | | | | | | | × | | |
| MAP sensor | × | | × | | | | | | | | | |
| Boost air temperature sensor | | | × | | | | | | | | | |
| MAF sensor | × | | × | | | | | | | | | |
| IAT sensor No.1 | × | | × | | | | × | | | × | | |
| IAT sensor No.2 | × | × | × | | | | | | | | | |
| ECT sensor | × | × | × | × | × | × | | | | × | | × |
| Fuel temperature sensor | | | | | | | | | | | | |
| APP sensor No. 1, No.2 | × | | × | × | | | | | | × | | × |
| Fuel pressure sensor | | | × | | | | | | | × | | |
| Intake shutter valve position sensor | | | | | | | | | | | | |
| Exhaust shutter valve position sensor | | | × | | | | | | | | | |
| Coolant control valve position sensor | | | | | | | | | | | | |
| HP-EGR control valve position sensor | × | | × | | | | | | | | | |
| LP-EGR control valve position sensor | × | | × | | | | | | | | | |
| Exhaust gas pressure sensor No.1 | × | | × | | | | | | | | | |
| Exhaust gas pressure sensor No.2 | | | × | | | | | | | | | |
| BARO sensor | × | | × | | | | | | | × | | |
| Battery | | × | | | × | | × | | | × | | × |
| Refrigerant pressure sensor | | | | × | × | | | | | | | |
| Exhaust gas temperature sensor No.1 | × | | × | | | | | | | | | |

| Item | EGR CONTROL | AIR FUEL RATIO (A/F) SENSOR HEATER CONTROL | DIESEL PARTICULATE FILTER REGENERATION CONTROL | A/C CUT-OFF CONTROL | ELECTRICAL FAN CONTROL | STARTER CUT-OFF CONTROL | GENERATOR CONTROL | RUN DRY PREVENTION (RDP) CONTROL | BLOW-BY HEATER CONTROL | i-stop CONTROL | DC-DC CONVERTER CONTROL | ENGINE OIL CONTROL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Exhaust gas temperature sensor No.2 | | | × | | | | | | | | | |
| Exhaust gas temperature sensor No.3 | | | × | | × | | | | | | | |
| Exhaust gas temperature sensor No.4 | × | × | × | | | | | | | | | |
| Engine oil temperature sensor | | | × | | | | | | | | | × |
| Engine oil pressure sensor | × | | × | | | | | | | | | |
| Actuator position sensor (turbocharger with variable turbine geometry) | | | × | | | | | | | | | |
| Speed sensor assy (turbocharger with variable turbine geometry) | | | | | | | | | | | | |
| Ambient temperature sensor | | | | | | | | | × | | | |
| A/F sensor | × | | × | | | | | | | | | |
| Power brake unit vacuum sensor | | | | | | | | | | × | | |
| CKP sensor | × | × | × | × | | × | × | | × | × | | × |
| CMP sensor | × | | | | | | | | | × | | |
| Back-up light switch (MTX) | | | | | | | | | | × | | |
| Neutral switch No.1 (MTX) | | | × | × | | × | | | | × | | |
| CPP switch (MTX) | × | | × | | | | | | | × | | |
| Starter interlock switch (MTX) | | | | | | × | | | | × | | |
| Neutral switch No.2 (MTX) | | | | | | | | | | × | | |
| Clutch stroke sensor (MTX) | | | | | | | | | | × | | |
| Generator (stator coil) (without i-ELOOP) | | | | | | | × | | | | | |
| CAN*1 | × | | × | × | × | × | × | × | × | × | × | × |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|
| LIN*2 | | | | | | | | | | | | |
| Output device | | | | | | | | | | | | |
| Intake shutter valve | | | × | | | | | | | × | | |
| Exhaust shutter valve | | | | | | | | | | × | | |
| HP-EGR control valve | × | | × | | | | | | | × | | |
| LP-EGR control valve | × | | × | | | | | | | × | | |
| Fuel pressure relief valve | | | | | | | | | | | | |
| Main relay | | | | | | | | | | | | |
| Turbocharger solenoid valve | | | × | | | | | | | | | |
| A/F sensor heater | | × | | | | | | | | | | |
| Starter relay | | | | | | × | | | | × | | |
| DC-DC converter (without i-ELOOP) | | | | | | | | | | | × | |
| Generator (Field coil) (without i-ELOOP) | | | | | | | × | | | × | | |
| A/C relay | | | | × | | | | | | × | | |
| Fan control module | | | | | × | | | | | | | |
| Cooling fan relay | | | | | × | | | | | | | |
| Engine oil solenoid valve | | | × | | | | | | | | | × |
| Suction control valve | | | | | | | | | | | | |
| Fuel injector No. 1—No.4 | | | × | | | | | × | | × | | |
| Blow-by heater relay | | | | | | | | | × | | | |
| Electric water pump | | | | | | | | | | | | |
| Coolant control valve | | | | | | | | | | | | |
| LIN*2 | | | × | | | | | | | × | | |
| CAN*3 | | | × | | | | × | | × | × | | |

*1 : DSC HU/CM, TCM (ATX), EPS control module, SAS control module, instrument cluster, BCM, start stop unit

*2 : Glow control module, engine oil level sensor

*3 : Instrument cluster (charging system warning light, master warning light, glow indicator light, diesel particulate filter indicator light, i-stop warning light (amber)/i-stop indicator light (green), i-stop indicator alarm/i-stop warning alarm)